

Randomness Requirements for Oblivious Routing on the Hypercube

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Abstract

We study the role of randomness in oblivious permutation routing on the n -dimensional hypercube. Our focus is on understanding how the number of random bits used by a routing algorithm affects its performance.

We first discuss known lower bounds showing that any oblivious routing algorithm achieving expected routing time $O(n)$ must use $\Omega(n)$ random bits. We then present an existential argument showing that $O(n)$ random bits suffice to achieve $O(n)$ expected routing time. Finally, we describe a refined time–randomness tradeoff which shows that reducing randomness further leads to a degradation in the probability of completing routing within a given time bound.

1 Introduction

Permutation routing on the hypercube is a fundamental problem in parallel computation. In this setting, each node initially holds a packet, and each packet has a unique destination. The goal is to route all packets to their destinations as efficiently as possible.

A natural class of algorithms for this problem is *oblivious routing*, where the path taken by each packet depends only on its source and destination. While such algorithms are simple and scalable, deterministic oblivious routing schemes can perform poorly in the worst case due to congestion.

Randomization provides a way to overcome these limitations by distributing traffic more evenly across the network. However, the amount of randomness required to achieve efficient routing is not immediately clear. This leads to the central question studied in this paper:

How many random bits are necessary and

sufficient to achieve $O(n)$ expected routing time in oblivious routing on the hypercube?

We address this question by examining both lower and upper bounds, and by discussing a fundamental tradeoff between randomness and routing performance.

2 Model and Preliminaries

We consider the n -dimensional hypercube with

$$N = 2^n$$

nodes. Each node is labeled with an n -bit binary string, and two nodes are adjacent if their labels differ in exactly one bit.

In the permutation routing problem, each node initially holds one packet, and each packet has a unique destination node. Thus, the destinations define a permutation on the set of nodes.

An oblivious routing algorithm specifies, for every source–destination pair, a fixed path along which the corresponding packet will be routed. The routing process proceeds in discrete time steps, where at each step a packet may traverse one edge.

We evaluate a routing algorithm using the following measures:

- **Routing time:** The number of steps required until all packets reach their destinations.
- **Randomness:** The total number of random bits used by the algorithm.

Throughout this paper, we focus on randomized oblivious routing algorithms and measure performance in terms of the expected routing time.

3 Problem Statement

Our goal is to understand the relationship between randomness and routing efficiency. In particular, we are interested in determining whether there exist constants $c_1, c_2 > 0$ such that an oblivious routing algorithm can satisfy

$$R \leq c_1 n \quad \text{and} \quad \mathbb{E}[\text{routing time}] \leq c_2 n,$$

where R denotes the number of random bits used.

We approach this question in three steps:

1. We show that achieving expected routing time $O(n)$ requires $\Omega(n)$ random bits.
2. We present an existential argument demonstrating that $O(n)$ random bits are sufficient.
3. We discuss a refined tradeoff showing that further reducing randomness affects the probability of completing routing within a fixed time bound.

4 Overview of Results

Our investigation leads to the following conclusions:

- Any oblivious routing algorithm achieving expected routing time $O(n)$ must use at least $\Omega(n)$ random bits.
- There exist randomized oblivious routing schemes that use $O(n)$ random bits and achieve expected routing time $O(n)$.
- A more detailed analysis reveals a tradeoff between randomness and reliability: reducing the number of random bits leads to a decrease in the probability of completing routing within a fixed time bound.

These results together show that $\Theta(n)$ random bits are both necessary and sufficient for achieving efficient oblivious routing on the hypercube.

5 Lower Bound on Randomness

We now show that achieving expected routing time $O(n)$ requires a linear number of random bits. The proof is based on a counting argument together with known lower bounds for deterministic oblivious routing.

5.1 Main Argument

Suppose a randomized oblivious routing algorithm uses at most k random bits. Then the algorithm induces a distribution over at most 2^k deterministic oblivious routing algorithms. Let these be denoted by

$$\mathcal{A}_1, \mathcal{A}_2, \dots, \mathcal{A}_{2^k}.$$

It is known (see Theorem 4.4 in [1]) that for any deterministic oblivious routing algorithm on the hypercube, there exists a permutation routing instance for which the routing time is at least

$$\Omega\left(\sqrt{\frac{N}{n}}\right),$$

due to congestion.

Using a standard averaging argument (Pigeonhole Principle), one can show that there exists a permutation π^* such that a non-negligible fraction of the algorithms $\mathcal{A}_i$ incur this lower bound on π^* . Consequently, for this permutation, the expected routing time of the randomized algorithm satisfies

$$\mathbb{E}[\text{routing time}] \geq \Omega\left(2^{-k} \cdot \sqrt{\frac{N}{n}}\right).$$

Substituting $N = 2^n$, we obtain

$$\mathbb{E}[\text{routing time}] \geq \Omega\left(2^{-k} \cdot \frac{2^{n/2}}{\sqrt{n}}\right).$$

In order for the expected routing time to be $O(n)$, it must hold that

$$2^{-k} \cdot \frac{2^{n/2}}{\sqrt{n}} \leq O(n).$$

Rearranging,

$$2^{-k} \cdot 2^{n/2} \leq O(n\sqrt{n}).$$

Taking logarithms, we obtain

$$-k + \frac{n}{2} \leq O(\log n),$$

which implies

$$k \geq \frac{n}{2} - O(\log n).$$

Thus, we conclude that

$$k = \Omega(n).$$

5.2 Conclusion

Any oblivious routing algorithm that achieves expected routing time $O(n)$ must use at least $\Omega(n)$ random bits. This establishes that linear randomness is necessary for efficient oblivious routing on the hypercube.

6 Existence of Low-Randomness Routing Schemes

We now show that $O(n)$ random bits suffice to achieve $O(n)$ expected routing time. The argument is based on the probabilistic method (see Theorem 5.10 in [1]).

Consider the family of all deterministic oblivious routing algorithms obtained by fixing the intermediate destinations in the standard randomized routing scheme. Since each of the $N = 2^n$ packets may choose any of N intermediate nodes, there are N^N such deterministic algorithms. Let us denote these algorithms by $A_1, A_2, \dots, A_{N^N}$.

For any fixed permutation routing instance, it is known that at most a $1/N$ fraction of these algorithms require more than $14n$ steps to complete routing. We refer to such algorithms as *bad* for that instance.

We construct a smaller family by sampling N^3 algorithms independently and uniformly at random (with replacement) from the set $\{A_1, \dots, A_{N^N}\}$. Let B denote the resulting multiset of sampled algorithms.

Fix a permutation routing instance. Since each sampled algorithm is bad with probability at most $1/N$, the expected number of bad algorithms in B is at most N^2 . By a Chernoff bound, the probability that more than $2N^2$ sampled algorithms are bad is at most $\exp(-N^2/4)$.

Now consider all $N!$ permutation routing instances. By a union bound, the probability that there exists any instance for which more than $2N^2$ algorithms in B are bad is at most

$$N! \cdot \exp(-N^2/4),$$

which is less than 1 for sufficiently large N . Therefore, there exists a choice of B such that, for every permutation routing instance, at most $2N^2$ algorithms in B are bad.

Fix such a set B . For any instance, selecting an algorithm uniformly at random from B results in a bad outcome with probability at most

$$\frac{2N^2}{N^3} = \frac{2}{N}.$$

Thus, with probability at least $1 - 2/N$, routing completes within $14n$ steps. It follows that the expected routing time is at most $15n$.

Since $|B| = N^3$, selecting an algorithm from B requires only $\log(N^3) = O(n)$ random bits. This establishes the existence of an oblivious routing scheme that uses $O(n)$ random bits and achieves $O(n)$ expected routing time.

6.1 Conclusion

The above argument establishes that there exist constants $c_1, c_2 > 0$ such that an oblivious routing scheme on the n -dimensional hypercube can be implemented using at most $c_1 n$ random bits while achieving expected routing time at most $c_2 n$. In particular, there exists a family B of deterministic routing algorithms of size $2^{O(n)}$ such that selecting an algorithm uniformly at random from B yields linear expected routing time for every permutation instance.

We note that the construction is non-uniform: while the probabilistic method guarantees the existence of such a family B , it does not provide an efficient procedure for constructing B given N .

7 Tradeoff Between Randomness and Success Probability

In addition to the existential result that $\Theta(n)$ random bits are sufficient to achieve $O(n)$ expected routing time, prior work studies a finer tradeoff between randomness and routing reliability. There is also a third variable that changes when we change the amount of randomness and completion time. In particular, the paper by Krizanc, Peleg, and Upfal [2] analyzes oblivious routing using three parameters: the time bound T , the number of random bits R , and the success probability Q , where Q denotes the probability that routing does not complete within time T . Their result shows that reducing the amount of randomness comes at a cost in success probability, revealing a fundamental tradeoff between randomness and the likelihood of completing routing within a fixed time.

7.1 Approach and Main Result

Krizanc, Peleg, and Upfal approach oblivious routing from an information-theoretic perspective by simultaneously analyzing three parameters: the time bound T , the failure probability Q , and the amount of randomness R , measured as the entropy of the random source used by the algorithm. Since an oblivious routing algorithm must fix its routing decisions without adapting to the evolving traffic pattern, randomness plays a crucial role in distributing packets across the network and avoiding congestion. The central question they address is how much randomness is required to guarantee that routing completes within T steps with high probability.

Their analysis proceeds by deriving a lower bound on the amount of randomness required to achieve a given performance guarantee. Using entropy arguments, they show that any oblivious routing algorithm that routes an arbitrary permutation within T steps with success probability $1 - Q$ must use at least

$$(1 - Q) \frac{\log N}{T} - \log Q - O(1)$$

random bits. Intuitively, if the routing must complete quickly and with high reliability, then the algorithm must inject sufficient randomness to ensure that packets are well distributed across the network and do not create bottlenecks.

In addition to this lower bound, they show that the tradeoff is essentially tight. They construct oblivious routing schemes whose randomness matches this bound up to lower-order terms, thereby demonstrating that the dependence between T , R , and Q cannot be significantly improved. They also provide an explicit family of algorithms whose randomness is within a logarithmic factor of the optimal for the same routing time. These results together establish a near-optimal characterization of the tradeoff between randomness and routing performance.

7.2 Illustration of the Randomness–Reliability Tradeoff

To better understand the relationship between randomness and routing reliability, we illustrate the tradeoff using the theoretical relationship between the number of random bits R , the time bound T , and the failure probability Q . For concreteness, we consider a hypercube of dimension $n = 20$ (i.e., $N = 2^{20}$ nodes). For fixed values of T , we plot the success

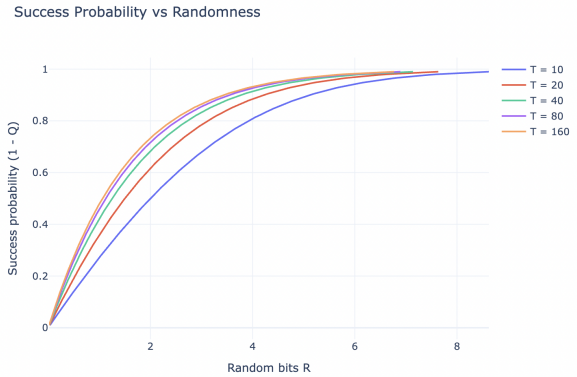


Figure 1: Success probability as a function of randomness for different time bounds T on a hypercube with $N = 2^{20}$ nodes. Increasing randomness improves reliability, while larger time bounds allow higher success probability for the same randomness.

probability $1 - Q$ as a function of the available randomness. Each curve corresponds to a different time bound, and demonstrates how the success probability improves as more randomness is introduced. The curves also highlight that larger time bounds allow the algorithm to achieve higher success probabilities for the same amount of randomness, since the routing process has more time to resolve congestion.

The figure clearly illustrates the tradeoff between randomness and reliability. For small values of R , the success probability is low, indicating that insufficient randomness leads to congestion and routing failures within the given time bound. As R increases, the success probability improves rapidly, eventually approaching 1. Additionally, for larger values of T , the curves lie above those for smaller T , showing that increasing the allowed routing time reduces the dependence on randomness. This behavior is consistent with the theoretical result that limiting randomness necessarily decreases the probability of completing routing within a fixed deadline.

8 Conclusion

In this paper, we studied the role of randomness in oblivious routing on the n -dimensional hypercube. We first discussed known lower bounds showing that any oblivious routing algorithm with expected routing time $O(n)$ must use $\Omega(n)$ random bits. We then presented an existential argument showing that

$O(n)$ random bits are sufficient to achieve $O(n)$ expected routing time. It is worth noting that this existence proof is non-uniform: while it guarantees the existence of a good family of routing schemes, it does not provide an explicit efficient procedure for constructing such a family given N .

We also considered a refined time–randomness tradeoff from prior work, which shows that reducing the number of random bits further leads to a decrease in the probability of completing routing within a fixed time bound. Together, these results show that there is a fundamental balance between randomness, routing time, and reliability in oblivious routing. In particular, $\Theta(n)$ random bits are both necessary and sufficient for efficient oblivious routing on the hypercube.

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References

- [1] R. Motwani and P. Raghavan, *Randomized Algorithms*, Cambridge University Press, 1995.
- [2] D. Krizanc, D. Peleg, and E. Upfal, “A Time-Randomness Tradeoff for Oblivious Routing,” *SIAM Journal on Computing*, 23(4):826–847, 1994.
- [3] S. Har-Peled, “Probabilistic Method II (Lecture Notes),” 2013.